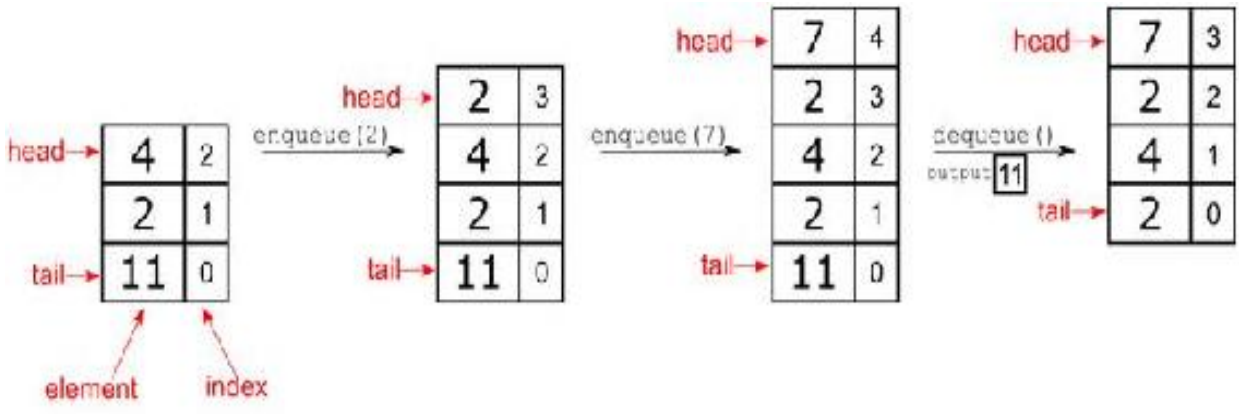


Figure D.3 Operations on a queue. Insertion (enqueue) adds elements to front of the queue. Delete (dequeue) remove the element at the back of the queue and returns it.

D.4 A Comparative Analysis for Containers

In summary, we can partition the abstract structures discussed according to a few key characteristics: if keys are maintained in a certain order, if they need to be unique, and if the container is associative. In the analysis provided in table D.1 we include all containers shown above and dictionaries, discussed in appendix C.

Table D.1 Comparison of containers

Structure	Order	Unique	Associativity
Bag	No	No	None
Stack	Yes	No	None